

Xiaoyu Zhang

CONTACT INFORMATION

340 Davis Hall
Department of Computer Science & Engineering
University at Buffalo, SUNY
Buffalo, NY, 14260-2500 USA

Phone: (716) 907-7845
E-mail: zhang376@buffalo.edu
Page: <https://zxy340.github.io/>

RESEARCH SUMMARY

I build **multimodal sensing agents** that *perceive, reason, and act* on the physical world by jointly leveraging sensing modalities organized across four complementary physical domains: **radio-frequency** (mmWave FMCW radar, synthetic-aperture and cascaded-array imaging), **acoustic / ultrasonic** (photoacoustic imaging, ultrasound, audio and speech), **optical / radiographic** (visible-light vision, infrared, mammography), and **bio-electrical** (wearable ear-EEG). My research closes the loop from **novel sensing front-ends** through **physics-informed inverse imaging** and **cross-modal representation learning** to **deployed clinical systems**, with validated impact on **smart health** (wound care, breast cancer subtyping, thyroid diagnosis, Alzheimer’s screening, hearing disorders) and **social good** (sustainable building, accessible HCI, point-of-care diagnostics). My work has produced **14 peer-reviewed papers** (7 first-authored, at venues including ACM IMWUT, ACM UIST, IEEE IoTJ, IEEE/RSJ IROS, and npj Imaging), a **Best Paper Candidate** at ACM SenSys Human-Sys’25, **4 patents** (one being translated through a startup I co-founded, semifinalist at Panasci TEC 2025), and a **clinical station deployed under an IRB-approved validation study**, reflecting a deliberate *research–clinic–industry* translation pipeline.

RESEARCH INTERESTS

My research lies at the intersection of **Wireless Sensing, Multimodal Medical AI, and Embodied Intelligence**. One technical question recurs across all three: *when the sensor is cheap and the data are scarce, where should the burden of perception sit — on physics, on learning, or on the hardware aperture?*

1) mmWave Sensing and Imaging. *Core direction of my dissertation.* I push commodity mmWave radar from coarse region-level sensing to millimeter-scale imaging of tissue and human motion, distributing the burden three ways: on *physics* — embedding Fresnel reflection and dielectric-mixing laws in the estimator so moisture maps are inverted rather than learned (sub-1% error through gauze, mmSkin, IoTJ’25); on *learning* — a depth teacher decouples position from pose and size, giving pixel-level hand localization from a single device (mmHand, IoTJ’25); and on *hardware* — a cascaded 12 Tx × 16 Rx array collapses a dense 2D scan into a sub-minute sweep (WaveSight-RF). Also covers mmWave–voice identification (UIST’23), micro-expression sensing, and privacy-preserving radar monitoring.

2) Multimodal Medical AI. I build diagnostic models that fuse and transfer across imaging and physiological modalities under scarce labels: tri-modal photoacoustic / ultrasound / mammographic fusion for non-invasive breast-cancer molecular subtyping, attention-alignment distillation for unified thyroid nodule diagnosis and segmentation, radiomics and deep-feature models for photoacoustic foot-ulcer prognosis (npj Imaging’26, Photoacoustics’25), and speech foundation models for Alzheimer’s subtype screening (ICIBM’26).

3) Robotic Perception and Autonomous Inspection. Aimed at sustainable construction and building energy efficiency, I carry the same sensing-to-decision stack outside the clinic, where the decision-maker is a factory operator or energy auditor rather than a physician: autonomous robotic inspection of building thermal insulation through subsurface sensing (IROS’26), and closed-loop vision-based process control for sustainable construction-material production. Two of my four filed patents originate in this direction.

EDUCATION

University at Buffalo, the State University of New York (SUNY) Sep. 2021 - Jul. 2026
 Ph.D., Computer Science and Engineering Co-advised by Prof. Wenyao Xu & Yaxiong Xie
 Dissertation (**defended Jul. 2026**): *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications*
 Committee: Wenyao Xu, Yaxiong Xie, Jun Xia

University of Science and Technology of China Sep. 2017 - Jun. 2020
 M.S., Electronics and Communication Engineering Advised by Prof. Bin Liu

Hefei University of Technology Sep. 2013 - Sep. 2017
 B.Eng., Electronic Information Engineering

PUBLICATIONS

Summary. 14 peer-reviewed papers, 7 first-authored (name in **bold**), spanning flagship venues in ubiquitous computing (ACM IMWUT, UIST), robotics (IEEE/RSJ IROS), medical imaging (Photoacoustics, npj Imaging), and biomedical informatics (IEEE JBHI, IoTJ). Thrust tags: [\[Health\]](#) [\[Embodied\]](#) [\[Social Good\]](#).

- [IROS'26] [\[Social Good\]](#) **Xiaoyu Zhang**, Ye Zhan, Chenhan Xu, Zhengxiong Li, Chuqin Huang, Yanda Cheng, Licheng Liang, Wei Bo, Jun Xia, Chi Zhou, Hongyue Sun, Wenyao Xu, “*Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing*”, IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. (*Patent filed.*)
- [IMWUT'26] [\[Health\]](#) Shuwei Hou, Wei Bo, L. Cao, Varun Shijo, Chuhui Liu, **Xiaoyu Zhang**, L. Guo, Wenyao Xu, “*Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis*”, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2026.
- [ICIBM'26] [\[Health\]](#) **Xiaoyu Zhang**, Wei Bo, Anarghya Das, Longxiang Pan, Wenyao Xu, “*Matching Speech Models to AD/HD Subtypes: A Mechanistic Benchmark*”, International Conference on Intelligent Biology and Medicine, 2026.
- [npj Imaging'26] [\[Health\]](#) Yanda Cheng, Chuqin Huang, Shuliang Yu, Saptarshi Chakraborty, Yunqi Xi, Robert W. Bing, Huijuan Zhang, **Xiaoyu Zhang**, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, “*Predictive modeling of chronic foot ulcer outcomes using longitudinal photoacoustic imaging*”, npj Imaging, 2026.
- [IoTJ'25] [\[Health\]](#) **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “*mmSkin: An Over-gauze Wound Assessment System using Radio Frequency Technologies*”, IEEE Internet of Things Journal, 2025. (*Translated into US patent; being commercialized through a co-founded startup.*)
- [IoTJ'25] [\[Embodied\]](#) **Xiaoyu Zhang**, Zhengxiong Li, Chenhan Xu, Luchuan Song, Huining Li, Hongfei Xue, Yingxiao Wu, Wenyao Xu, “*mmHand: Towards Pixel-Level-Accuracy Hand Localization Using a Single Commodity mmWave Device*”, IEEE Internet of Things Journal, 2025.
- [HumanSys'25] [\[Health\]](#) **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “*Through-dressing Wound Monitoring Based on the mmWave Sensor*”, The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM SenSys), 2025. (**Best Paper Candidate**)
- [Sensors'25] [\[Health\]](#) **Xiaoyu Zhang**, Chuhui Liu, Yanda Cheng, Zhengxiong Li, Chenhan Xu, Chuqin Huang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “*A Comprehensive Survey of Research Trends in mmWave Technologies for Medical Applications*”, Sensors, 2025.

- [Photoacoustics’25] [Health] Chuqin Huang, Yanda Cheng, **Xiaoyu Zhang**, Ye Zhan, Wenhan Zheng, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, “*Radiomics-Driven Perfusion Prediction in Clinical Photoacoustic Foot Imaging*”, Photoacoustics, 2025.
- [Photoacoustics’25] [Health] Chuqin Huang, Emily Zheng, Wenhan Zheng, Huijuan Zhang, Yanda Cheng, **Xiaoyu Zhang**, Varun Shijo, Robert W. Bing, Isabel Komornicki, Linda M. Harris, Ermelinda Bonaccio, Kazuaki Takabe, Emma Zhang, Wenyao Xu, Jun Xia, “*Enhanced clinical photoacoustic vascular imaging through a skin localization network and adaptive weighting*”, Photoacoustics, 2025.
- [JBHI’24] [Health] Wei Bo, Suzanne S. Sullivan, **Xiaoyu Zhang**, Mingchen Gao, Wenyao Xu, “*A Telemedicine Analytic Framework for Fully and Semi-automatic Alzheimer’s Disease Screening using Clock Drawing Test*”, IEEE Journal of Biomedical and Health Informatics, 2024.
- [UIST’23] [Embodied] Tiantian Liu, Feng Lin, Chao Wang, Chenhan Xu, **Xiaoyu Zhang**, Zhengxiong Li, Wenyao Xu, Ming-Chun Huang, Kui Ren, “*WaveID: Robust and Secure Multi-modal User Identification via mmWave-voice Mechanism*”, ACM Symposium on User Interface Software and Technology, San Francisco, USA, 2023.
- [BSN’19] [Embodied] **Xiaoyu Zhang**, Bin Liu, “*A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs*”, IEEE-EMBS International Conference on Body Sensor Networks, Chicago, USA, 2019.
- [Bodynets’17] [Embodied] Chengjie Guan, Bin Liu, Zhiqiang Liu, Y. Zhang, **Xiaoyu Zhang**, “*JMMM: A Mobility Model for WBANs Based on Human Joint Movements*”, 19th EAI International Conference on Body Area Networks, Dalian, China, 2017.

HONORS AND AWARDS

- **Best Paper Candidate Award**, HumanSys’25 — The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (co-located with ACM SenSys), 2025
- **Semifinalist**, Panasci Technology Entrepreneurship Competition (Panasci TEC), University at Buffalo, 2025
- **Graduate Teaching Award**, Department of CSE, University at Buffalo, 2024
- **Chair’s Fellowship**, Department of CSE, University at Buffalo, 2021
- Merit scholarships (graduate and undergraduate), USTC and HFUT, 2015–2019

INDUSTRY & CLINICAL EXPERIENCE

Clinical Research Collaborations

2022 - Present

Multi-year IRB-approved clinical collaborations at the University at Buffalo and partner sites, spanning device deployment, on-site data collection, and multi-modal cohort curation:

- **Wound care** (UBMD Vascular Surgery) — deployed the WaveSight-RF clinical station on site; **5 patients** scanned to date under an IRB-approved protocol, with repeat longitudinal scanning up to **3 months** per patient
- **Breast imaging** — clinical cohort of **284 imaged patients**; curated the **53-patient** tri-modal photoacoustic / ultrasound / mammography subset used for molecular subtyping
- **Wearable ear-EEG** — **27 subjects** recorded across four passive auditory tasks (~1,800 trials), **23** retained after per-channel quality control, paired with standardized clinical score scales
- **Thyroid ultrasound** — ongoing clinical scanning collaboration for nodule diagnosis and lesion segmentation
- **Photoacoustic Imaging Lab** (Dept. of Biomedical Engineering) — joint imaging-system development and radiomics analysis

Also curated and quality-filtered a public speech corpus for Alzheimer’s-subtype benchmarking. Responsibilities across these collaborations: IRB protocol support, on-site device operation and data collection, multi-modal dataset curation, and clinician-facing decision-support design.

China Merchants Bank Software Center

Jul. 2020 - Sep. 2020

Designed and implemented a full-stack data-verification system (Bootstrap + Django) with dynamic front-back interaction for multi-source financial data.

COMPETITION

- **Panasci Technology Entrepreneurship Competition (Panasci TEC) 2025** Mar. 2025
Advanced Through-Gauze Wound Assessment Technology.
Xiaoyu Zhang (technical lead / co-founder), Chuhui Liu, Jun Xia, Linda M. Harris, Wen Yao Xu.
Advanced to the **semifinals**; pitched the patented mmWave wound-assessment platform to clinical and industry judges.

PRESENTATIONS & DEMOS

- **Ph.D. Dissertation Defense:** *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications* Jul. 2026
University at Buffalo, CSE
- **Live Demonstration:** SUNY Research Expo Mar. 2026
WaveSight-RF clinical wound-imaging station — translational outreach
- **Community Engagement Studio:** UB CTSI Feb. 2026
Wound-measurement device design review with a 12-member community expert panel
- **Invited Talk:** The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys, co-located with ACM SenSys) May 2025
Through-dressing Wound Monitoring Based on the mmWave Sensor
- **Pitch Talk:** Panasci Technology Entrepreneurship Competition Mar. 2025
Advanced Through-Gauze Wound Assessment Technology
- **Invited Talk:** IEEE-EMBS International Conference on Body Sensor Networks May 2019
A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs
- **Invited Talk:** 19th EAI International Conference on Body Area Networks Sep. 2017
JMMM: A Mobility Model for WBANs Based on Human Joint Movements

TEACHING EXPERIENCES

Teaching Assistant across **6 semesters** at the University at Buffalo, including one invited guest lecture, plus one semester at USTC.

1. Algorithm Analysis and Design, UB Fall 2021, Spring 2022, Fall 2022, Fall 2023, Fall 2024
2. Special Topics, UB — incl. **invited guest lecture:** *Wireless Signal Processing: Making Sense of the Invisible* Fall 2023
3. Mathematical Logic and Graph Theory, USTC Fall 2018

MENTORING EXPERIENCES

I have mentored **10+ students** on wireless sensing, embedded systems, and AI-based medical screening projects.

- Prakriti Setlur — AI-based clinical image screening pipelines
- Kh Shafin Farhan — foundation models for medical imaging
- Abhi Ramtel (CSE@UB) — UI design of an over-gauze wound assessment system
- Cole Desimone (AE@UB) — 3D model design of the mmWave sensor scanning system
- George Gillman (EE@UB) — mmWave technologies for medical applications: a review
- Weida Jiang (CSE@UB) — embedded mmWave-based hand detection on Raspberry Pi
- Wenxuan Huang & Yiwen Tan (CSE@UB) — tooling for automatic image labeling
- High-school summer research interns — AI-based medical screening pipelines, mentored through UB outreach programs

Conference Organization:

- **Organizing Committee Member (Local Arrangements)**, International Conference on Intelligent Biology and Medicine (ICIBM) [2026]

Journal Reviewer:

- IEEE Transactions on Mobile Computing (TMC) [2025, 2026]
- ACM Transactions on Computing for Healthcare (HEALTH) [2025]
- IEEE Open Journal of Engineering in Medicine and Biology (OJEMB) [2025]
- Scientific Reports [2025]
- Smart Health (SH) [2024, 2025]

Conference Reviewer:

- ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) [2025]
- IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) [2024, 2026]
- Physics Education Research Conference (PERC), American Association of Physics Teachers (AAPT) [2026]
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) [2023, 2024]

Outreach:

- Live technology demonstration, **SUNY Research Expo** — translational outreach for the WaveSight-RF clinical wound-imaging station [Mar. 2026]
- **Community Engagement Studio**, UB Clinical and Translational Science Institute (CTSI) — convened a **12-member community expert panel** (healthcare providers, clinical trainees, community advocates) for a design review of the wound-measurement device [Feb. 2026]